

Classifying Kidney Transplant Rejection from Biopsy RNA

Transplant rejection and kidney biopsy

- **Rejection:** The recipient's immune system attacks the donated kidney.
- **Kidney biopsy:** A small tissue sample collected from the kidney for examination.
 - **Histology:** Microscopic examination reveals injury and inflammation that can support a rejection diagnosis.
 - **Molecular measurements:** Counts of selected RNA types reflect gene activity and the mix of cells in the tissue.

Possible Use: Molecular Second Opinion for Ambiguous Biopsies

- Sometimes the histology findings are ambiguous.

Example:

A biopsy shows mild inflammation, but not enough to diagnose rejection. In a study of 146 borderline diagnoses, inflammation disappeared in some patients. Others later developed acute rejection.

- As a step towards potentially helping resolve ambiguous biopsies, I used RNA counts to classify recorded diagnoses. A high rejection score could add evidence for specialist review.

Evidence for a molecular second opinion

Banff guidance 2026

Considers validated molecular tests for difficult biopsy interpretations

B-HOT study Rosales, 2022

Higher initial molecular scores in patients who later developed chronic active antibody-mediated rejection, despite no initial diagnosis of it

Why both kinds of error matter

Missed rejection can leave kidney injury untreated.
Unnecessary treatment can weaken the body's defenses and worsen an infection.

KDIGO (2009); Nankivell (2019); BK polyomavirus consensus (2024)

**This project has not established usefulness in ambiguous biopsies.
That requires comparing specialist assessment with and without RNA scores.**

What the dataset contains

1,395 biopsy specimens, with one row per specimen in each table.

RNA counts

Column	Example
Specimen ID (key)	GSM6510425
IFNG	4
GUSB (reference)	260
Other RNA counts	...

1 : 1

Same ID

Specimen information

Column	Example
Specimen ID (key)	GSM6510425
Recorded diagnosis	No rejection
Study group	Discovery
Other details	...

- Includes 1,193 transplant biopsies and 202 native-kidney controls.
- The study does not document separation by patient or referring center between study groups.

Outcome: rejection versus no rejection

The model combines the three rejection categories in the public dataset.

Recorded diagnosis	Meaning	Model label
Antibody-mediated rejection	Antibodies: immune proteins	Rejection
T-cell-mediated rejection	T cells: immune cells	Rejection
Mixed rejection	Both rejection processes	Rejection
No rejection	No recorded rejection	No rejection

The authors excluded some diagnostic categories from the dataset.

The measurements used to build the features

770 RNA counts per specimen, one for each RNA type measured.

758

Model measurements

- Selected human and viral RNA measurements.
- Signals reflect gene activity and the mixture of cells in the biopsy.
- Normalized values become the model's inputs.

12

Housekeeping references

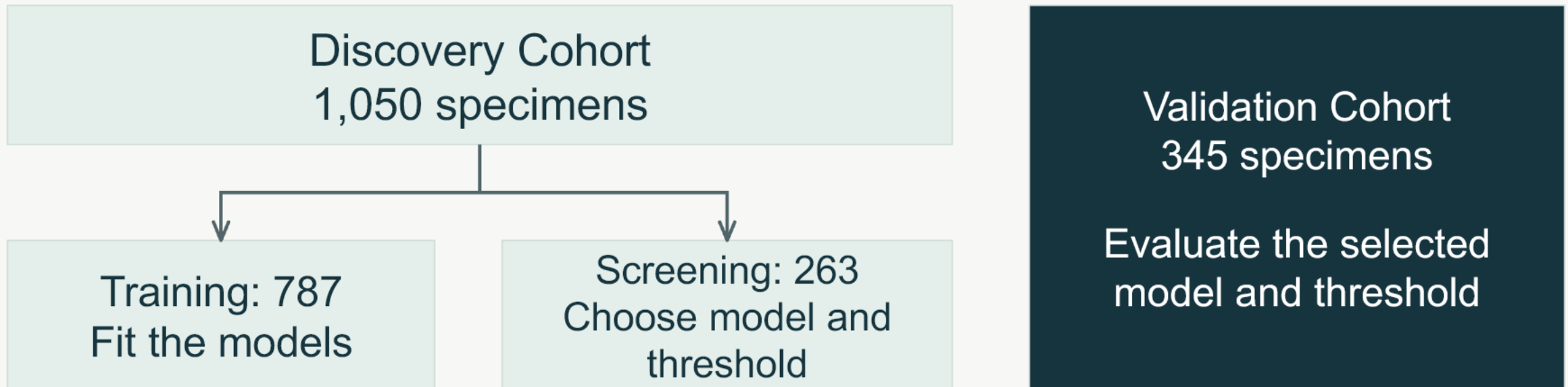
- Relatively stable RNAs provide a reference for each specimen.
- Help adjust for differences in overall measurable RNA input.
- Used for normalization, then excluded from the model.

Preparing counts for the model

For each of the 758 RNA counts	IFNG example
1. Start with the raw count	4
2. Calculate $\log_2(\text{count} + 1)$	2.322
3. Average $\log_2(\text{count} + 1)$ for the 12 reference RNAs	7.900
4. Subtract that reference	$2.322 - 7.900 = -5.578$

Development and evaluation specimens

- I used the authors' two study groups: discovery and validation cohorts.
- My training / screening split kept diagnosis proportions similar.



Models compared

Each model produces a score for rejection versus no rejection.

Model	Approach and rationale
Constant baseline	One score: the training rejection fraction
IFNG only (Logistic regression)	One immune-related RNA measurement as a simple benchmark
All RNA (Logistic Regression)	A weighted combination of 758 measurements. Regularization limits the size of the weights.
CatBoost	Can capture nonlinear RNA patterns and interactions. Shallow trees help limit overfitting.

Choosing a rejection threshold

Recall

Correctly flagged rejection cases

All cases diagnosed as rejection

(False negatives are missed cases.)

Precision

Correctly flagged rejection cases

All rejection flags

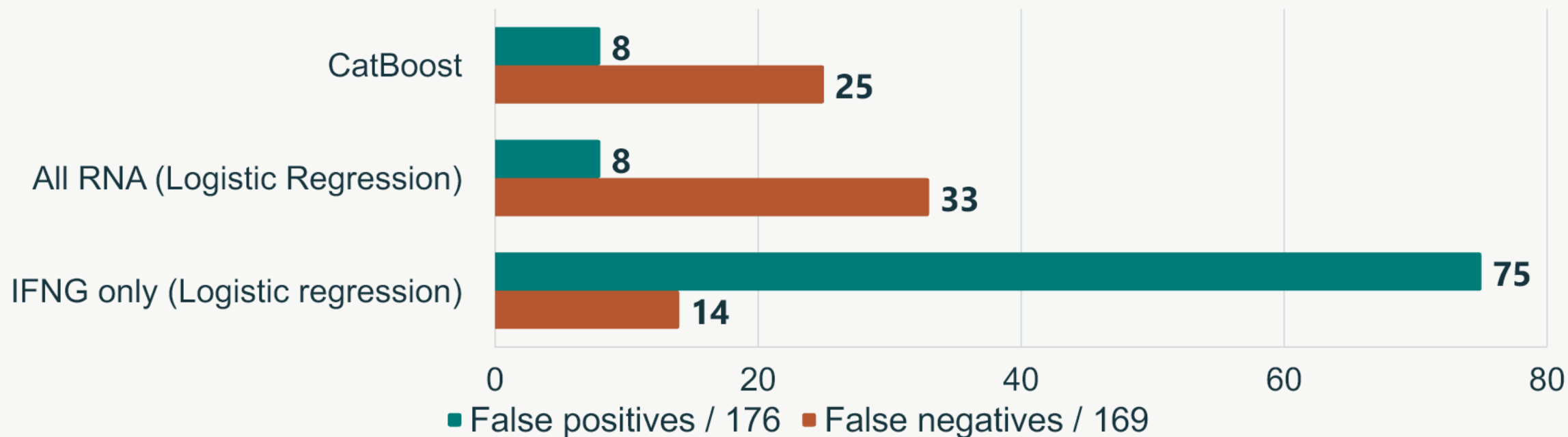
(False positives are incorrect flags.)

My selection rule for screening specimens

Step	Rule
1	Detect at least 90% of cases diagnosed as rejection.
2	Among qualifying choices, minimize false positives.
3	Keep the chosen model and threshold for evaluation.

Validation results

Same 345 specimens: 169 rejection and 176 no rejection



- Constant baseline: 0 false negatives, 176 false positives. It flags everyone.

Inspecting the analysis report

Saved analysis for another analyst to inspect

- Full model comparisons
- Errors by recorded diagnosis
- Score reliability and uncertainty

Live walkthrough of the preserved report

One public specimen through the service

- GSM6510425: recorded no rejection.

Model score	Selected threshold	Rejection flag
0.274934	0.876588	False (below cutoff)

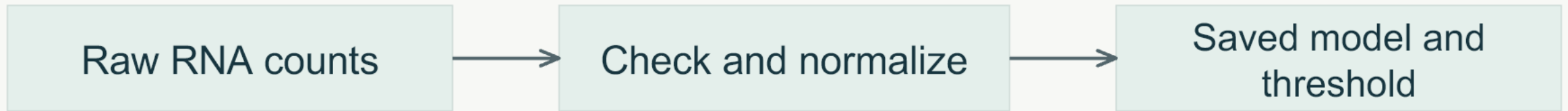
Model: 20260915_shared:any_rejection:catboost_all_depth4

- The complete input produces a flag that agrees with the recorded diagnosis.
- Removing IFNG clears the previous result and returns no new score.

“Missing required assay targets: IFNG”

Shared preparation and scoring

The analysis and application use the same preparation code.



Application entry point

```
def predict(self, counts: pd.DataFrame) -> pd.DataFrame:
    """Validate an entire raw batch before returning any specimen results."""
    normalized = normalize_counts(counts, self.schema)
    return self.predict_normalized(normalized)
```

The web page and command line both use this prediction path.

An interface other software can call

API: a documented way for software to send data and receive results.



Each result identifies the calculation

Specimen	Score	Threshold	Flag
GSM6510425	0.274934	0.876588	False

Each result includes versions for the model and expected input format.

Checking the application against the analysis

All 345 validation specimens

Saved results, command line and HTTP service had matching flags.
Scores agreed within 10^{-12} , including reordered measurement columns.

Example check: HTTP scores versus command-line scores

```
np.testing.assert_allclose(  
    actual.rejection_score, cli.rejection_score, rtol=0, atol=1e-12  
)
```

Failure checked	Expected response
Missing or duplicate measurements	No scores for the batch
Incompatible model metadata	Model does not load

Running and maintaining the application

What another developer receives	Why it helps
Locked environment and run instructions	Install the recorded dependencies and start the service.
Automated checks on code changes	Catch changes that break scoring or input handling.
Tested local container	Run the packaged application with its saved model.

Hosted checks passed for an earlier revision.

The current application and research container were checked locally.

Cloud deployment remains future work.

Sources: setup guide, automated-check workflow and dated verification records

What this project accomplished

A reproducible model comparison

Combined measurements reduced false flags, with a tradeoff in missed cases. The practical advantage over logistic regression remains uncertain.

Working software that preserves the calculation

A versioned score, clear input errors and checks against the saved results.

Added value for uncertain biopsies still needs direct evaluation.